L1: Adam, 500 epochs
Cells: mean ± SE. Bold: within one SE.Square: minimum. Circle: selected.

Hidden layers: (50,)

| Sigmoid Penalty strength λ | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.004 | 0.101 ±0.0039 | 0.101 ±0.0033 | 0.101 ±0.0037 |
| 0.001 | 0.1 ±0.004 | 0.0475 ±0.0075 | 0.0275 ±0.0079 | 0.0196 ±0.0043 |
| 0.0001 | 0.0305 ±0.0049 | 0.0192 ±0.0029 | **0.0173 ±0.004** | **0.0179 ±0.0042** |
| 1e-05 | 0.0279 ±0.0046 | **0.0178 ±0.003** | 0.0193 ±0.0049 | 0.02 ±0.0047 |

Hidden layers: (100, 100)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0038 | 0.1 ±0.0036 | 0.103 ±0.0035 | 0.102 ±0.0029 |
| 0.001 | 0.0982 ±0.0049 | 0.0999 ±0.0052 | 0.0993 ±0.0053 | 0.0979 ±0.0058 |
| 0.0001 | 0.0286 ±0.0049 | **0.0182 ±0.003** | 0.0192 ±0.0024 | **0.0175 ±0.0032** |
| 1e-05 | **0.0171 ±0.0042** | **0.0161 ±0.0029** | 0.0201 ±0.0039 | 0.0193 ±0.0032 |

Hidden layers: (150,)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.0999 ±0.0038 | 0.101 ±0.0034 | 0.104 ±0.0033 | 0.102 ±0.0053 |
| 0.001 | 0.0975 ±0.0047 | 0.0578 ±0.012 | 0.0277 ±0.0085 | 0.0197 ±0.003 |
| 0.0001 | 0.0975 ±0.005 | 0.0394 ±0.011 | 0.019 ±0.003 | 0.019 ±0.003 |
| 1e-05 | 0.0721 ±0.0077 | **0.0186 ±0.0031** | **0.0182 ±0.0021** | **0.0179 ±0.0032** |

Hidden layers: (300, 300)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0035 | 0.101 ±0.003 | 0.103 ±0.0041 | 0.107 ±0.0016 |
| 0.001 | 0.103 ±0.0061 | 0.111 ±0.0058 | 0.109 ±0.0074 | 0.165 ±0.034 |
| 0.0001 | 0.0826 ±0.0069 | 0.104 ±0.0081 | 0.0346 ±0.013 | 0.42 ±0.19 |
| 1e-05 | **0.0175 ±0.003** | 0.0236 ±0.0016 | 0.0268 ±0.0052 | 0.0796 ±0.021 |

Hidden layers: (50,)

| ReLU Penalty strength λ | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.102 ±0.0033 | 0.101 ±0.0046 |
| 0.001 | 0.0225 ±0.0035 | 0.0212 ±0.0038 | 0.0276 ±0.0034 | 0.0234 ±0.0048 |
| 0.0001 | 0.0219 ±0.0032 | 0.0202 ±0.0029 | **0.0194 ±0.0033** | **0.0195 ±0.0045** |
| 1e-05 | 0.0218 ±0.0033 | 0.0203 ±0.0037 | **0.018 ±0.0038** | **0.0193 ±0.0046** |

Hidden layers: (100, 100)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.102 ±0.0035 | 0.103 ±0.0033 |
| 0.001 | **0.0177 ±0.004** | **0.0175 ±0.0039** | 0.0211 ±0.004 | **0.0181 ±0.0037** |
| 0.0001 | 0.0199 ±0.0024 | **0.0194 ±0.0022** | 0.0216 ±0.0034 | **0.0185 ±0.0034** |
| 1e-05 | 0.0197 ±0.0027 | **0.0188 ±0.0026** | **0.0194 ±0.0039** | **0.0186 ±0.003** |

Hidden layers: (150,)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.102 ±0.0033 | 0.102 ±0.0037 |
| 0.001 | 0.022 ±0.0035 | 0.0224 ±0.0036 | 0.0235 ±0.0037 | 0.0197 ±0.0032 |
| 0.0001 | **0.0191 ±0.0029** | 0.0206 ±0.0033 | **0.0191 ±0.0032** | **0.0182 ±0.0034** |
| 1e-05 | 0.0209 ±0.0033 | 0.0205 ±0.0038 | **0.0182 ±0.0031** | **0.0194 ±0.0039** |

Hidden layers: (300, 300)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.101 ±0.0036 | 0.101 ±0.0038 |
| 0.001 | **0.0166 ±0.0029** | **0.0181 ±0.0038** | 0.0199 ±0.0038 | 0.0273 ±0.0053 |
| 0.0001 | **0.019 ±0.0032** | **0.0184 ±0.0032** | 0.0269 ±0.0055 | 0.0332 ±0.007 |
| 1e-05 | 0.0238 ±0.0041 | 0.0533 ±0.034 | 0.0233 ±0.003 | 0.0372 ±0.0048 |

Hidden layers: (50,)

| Leaky ReLU Penalty strength λ | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.102 ±0.0033 | 0.101 ±0.0046 |
| 0.001 | 0.0225 ±0.0036 | 0.0213 ±0.0038 | 0.0277 ±0.0035 | 0.0205 ±0.0038 |
| 0.0001 | 0.0219 ±0.0032 | 0.0201 ±0.0029 | **0.0192 ±0.0034** | **0.0187 ±0.004** |
| 1e-05 | 0.0218 ±0.0033 | 0.0204 ±0.0038 | **0.018 ±0.004** | 0.0197 ±0.0042 |

Hidden layers: (100, 100)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.102 ±0.0035 | 0.103 ±0.0033 |
| 0.001 | **0.0178 ±0.004** | **0.0177 ±0.0038** | 0.023 ±0.0049 | 0.0228 ±0.0038 |
| 0.0001 | 0.0197 ±0.0025 | 0.0198 ±0.0023 | 0.0199 ±0.0045 | **0.0187 ±0.0029** |
| 1e-05 | 0.0202 ±0.0024 | 0.0206 ±0.0025 | 0.0211 ±0.0029 | 0.0224 ±0.0034 |

Hidden layers: (150,)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.102 ±0.0033 | 0.102 ±0.0038 |
| 0.001 | 0.0221 ±0.0035 | 0.0225 ±0.0036 | 0.0232 ±0.0035 | 0.0197 ±0.0032 |
| 0.0001 | **0.0192 ±0.0029** | 0.0206 ±0.0033 | 0.0196 ±0.0031 | **0.0185 ±0.0035** |
| 1e-05 | 0.0208 ±0.0034 | 0.0197 ±0.0042 | **0.0177 ±0.0034** | **0.0187 ±0.0041** |

Hidden layers: (300, 300)

| | 0.003 | 0.01 | 0.05 | 0.1 |
|---|---|---|---|---|
| 0.1 | 0.1 ±0.0039 | 0.1 ±0.0039 | 0.101 ±0.0036 | 0.1 ±0.0039 |
| 0.001 | **0.0164 ±0.0029** | **0.0177 ±0.0036** | **0.0175 ±0.003** | 0.0229 ±0.0028 |
| 0.0001 | 0.0211 ±0.0029 | **0.0184 ±0.003** | 0.0217 ±0.0016 | 0.0264 ±0.0043 |
| 1e-05 | 0.0217 ±0.0033 | **0.0183 ±0.003** | 0.02 ±0.0024 | 0.0275 ±0.0046 |

Learning rate γ

Mean CV validation MSE
$4 \times 10^{-1}$
$3 \times 10^{-1}$
$2 \times 10^{-1}$
$10^{-1}$
$6 \times 10^{-2}$
$4 \times 10^{-2}$
$3 \times 10^{-2}$
$2 \times 10^{-2}$